

Age of The Drivables



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Road safety slogans like 'Alert Today; Alive Tomorrow' might become redundant in the future with the kind of safety, information, communication, entertainment and comfort features that are being built into today's cars.

We add to that fascination some lesser-known but equally exciting research projects, which are bound to augment the capabilities of tomorrow's smartcars.

Born-in-Bengaluru tech could help the visually impaired to drive

Since late last year, Tech Mahindra has incubated special goggles designed by its employee Kunal Bhat. Bhat, who accidentally banged his head on a wall in a dimly-lit staircase, decided to set about finding a tool that could help the visually impaired to visualise obstacles in their way much before these are sensed by their canes.

Working on this idea, he developed smartgoggles, which together with a smartphone, helps users to sense things on their path and form a 3D mental image of their surroundings.

How does it work? The smartphone camera captures the user's environment, identifies objects of collision and calculates their position and distance using special

algorithms. This information is wirelessly transmitted to the goggles. The smartglasses fitted in the goggles convey this information to the wearer through haptic feedback. It uses mild vibrations from relevant directions to help him or her feel the world around, for a radius of 1.5m (5-feet). A fluent flow of vibrations helps the user form a 3D mental image of the surroundings.

The current prototype is being tested by Tech Mahindra at Bengaluru with the help of two social organisations. Future versions are expected to have more capabilities like audio feedback, context-sensitive analysis, location and direction feedback, cloud based analysis, collective intelligence and inter-device communication.

Why this is featuring right at the beginning of a story on smartcars is because Tech Mahindra believes that this technology can be adapted to applications like autonomous cars and plans to upgrade the application to driverless cars in the future.

The mystic appeal of quantum tech to navigate into the future

The Global Navigation Satellite System (GNSS), which is being relied upon today for almost all forms of navigation, including that of smart and autonomous cars, is not

without its share of flaws. For one, it depends on signals sent to and from satellites launched into outer space, which makes it a costly affair to maintain the constellation and improve capacity.

Secondly, there are occasional reports from the US government agencies about GNSS-related security risks and their vulnerability to attack. Limitations of GNSSes under water are also known, making these ineffective for use with submarines.

In order to overcome all this, scientists at the UK

A man in an autonomous driving test vehicle

